

Introduction to Proof in Discrete Mathematics

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What is a Proof?

- ▶ A **proof** is a logical argument that verifies the truth of a mathematical statement.
- ▶ In discrete mathematics, proofs are used to validate statements about numbers, sets, graphs, and algorithms.

Types of Proof Techniques

There are several common techniques used in mathematical proofs:

1. Direct Proof

- ▶ Proves a statement by assuming the hypothesis and logically deriving the conclusion.
- ▶ Example: Proving that the sum of two even numbers is even.

2. Proof by Contradiction

- ▶ Assumes the negation of the statement and shows that this leads to a contradiction.
- ▶ Example: Proving that $\sqrt{2}$ is irrational.

3. Proof by Induction

- ▶ Used to prove statements about integers, typically involving sequences.
- ▶ Two steps: **Base Case** and **Inductive Step**.

4. Proof by Counterexample

- ▶ Demonstrates that a statement is false by providing a single example that contradicts it.

Direct Proof

- ▶ A **direct proof** demonstrates the truth of a statement by logically deriving the conclusion from the given information.
- ▶ It involves assuming the hypothesis is true and using logical steps to arrive at the conclusion.
- ▶ Commonly used for statements in the form:

If P , then Q .

Direct Proof: Example 1

Theorem: The sum of two even numbers is even.

Proof.

Let a and b be even numbers. We can express the even numbers as, there exist integers m and n such that:

$$a = 2m \quad \text{and} \quad b = 2n.$$

Then, the sum is:

$$a + b = 2m + 2n = 2(m + n).$$

Since $m + n$ is an integer, because the sum to two integers is and integer. So, $a + b$ is even. □

Direct Proof: Example 2

Theorem: The product of two odd numbers is odd.

Proof.

Let a and b be odd numbers. By definition of odd numbers, there exist integers m and n such that:

$$a = 2m + 1 \quad \text{and} \quad b = 2n + 1.$$

Then, the product is:

$$a \times b = (2m+1)(2n+1) = 4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1.$$

Since $2mn + m + n$ is an integer, $a \times b$ is odd. □

Direct Proof: Example 3

Theorem: The square of an even number is even.

Proof.

Let n be an even number. By definition, there exists an integer k such that:

$$n = 2k.$$

The square of n is:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, n^2 is even.



Direct Proof: Example 4

Theorem: The square of an odd number is odd.

Proof.

Let n be an odd number. By definition, there exists an integer k such that:

$$n = 2k + 1.$$

The square of n is:

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, $n^2 = 2m + 1$, where m is an integer. Thus, n^2 is odd. □

Summary of Direct Proof

Direct proof involves starting from known information (hypothesis) and applying logical reasoning to reach a conclusion.

Proof by Contraposition

- ▶ In a **proof by contraposition**, we prove a statement of the form:

If P , then Q

by proving its contrapositive:

If $\neg Q$, then $\neg P$.

- ▶ The contrapositive is logically equivalent to the original statement.
- ▶ This method is often easier than direct proof, especially when assuming Q false leads to a clearer argument.

Contraposition: Example 1

Theorem: If n^2 is even, then n is even.

Proof.

We will prove this by contraposition. The contrapositive of the statement is:

If n is odd, then n^2 is odd.

Assume n is odd. Then $n = 2k + 1$ for some integer k .

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, n^2 is odd. Thus, if n^2 is even, n must be even. □

Contraposition: Example 2

Theorem: If $x + y$ is odd, then one of x or y is odd.

Proof.

We will prove this by contraposition. The contrapositive of the statement is:

If both x and y are even, then $x + y$ is even.

Assume both x and y are even. Then, there exist integers m and n such that:

$$x = 2m \quad \text{and} \quad y = 2n.$$

Therefore:

$$x + y = 2m + 2n = 2(m + n).$$

Since $m + n$ is an integer, $x + y$ is even. Hence, if $x + y$ is odd, at least one of x or y must be odd. □

Contraposition: Example 3

Theorem: If n^2 is not divisible by 3, then n is not divisible by 3.

Proof.

We will prove this by contraposition. The contrapositive of the statement is:

If n is divisible by 3, then n^2 is not divisible by 3.

Assume n is divisible by 3. Then $n = 3k$ for some integer k .

$$n^2 = (3k)^2 = 9k^2 = 3(3k^2).$$

Since $3k^2$ is an integer, n^2 is divisible by 3. Thus, if n^2 is not divisible by 3, then n is not divisible by 3. □

Proof by Contradiction

Method: In a proof by contradiction, we assume the negation of the statement we want to prove and show that this assumption leads to a contradiction. This contradiction implies that our original assumption must be false, and therefore the statement we wanted to prove is true.

General Steps:

1. Assume that the statement to be proved is false.
2. Show that this assumption leads to a contradiction.
3. Conclude that the assumption must be false, and thus the original statement is true.

Contradiction: Example 1

Theorem: The sum of two odd numbers is even.

Proof:

- ▶ Assume, for the sake of contradiction, that the sum of two odd numbers is odd.
- ▶ Let the two odd numbers be $2m + 1$ and $2n + 1$, where m and n are integers.
- ▶ The sum of these two numbers is:

$$(2m + 1) + (2n + 1) = 2(m + n + 1).$$

This expression is clearly even, which contradicts the assumption that the sum is odd. Therefore, the sum of two odd numbers must be even.

Contradiction: Example 2

Theorem: The product of two even numbers is even.

Proof:

- ▶ Assume, for the sake of contradiction, that the product of two even numbers is odd.
- ▶ Let the two even numbers be $2m$ and $2n$, where m and n are integers.

The product of these two numbers is:

$$(2m)(2n) = 4mn.$$

Clearly, this is even because it is divisible by 2. This contradicts the assumption that the product is odd. Therefore, the product of two even numbers must be even.

Contradiction: Example 3

Theorem: The sum of an even number and an odd number is odd.

Proof:

- ▶ Assume, for the sake of contradiction, that the sum of an even number $2m$ and an odd number $2n + 1$ is even.

The sum is:

$$(2m) + (2n + 1) = 2(m + n) + 1.$$

This is clearly odd, not even. Therefore, our assumption is false.

Hence, the sum of an even number and an odd number is odd.

Contradiction: Example 4

Theorem: There is no integer x such that $x^2 = 3$.

Proof:

- ▶ Assume, for the sake of contradiction, that there is an integer x such that $x^2 = 3$.

Then $x^2 = 3$, and we need to check if any integer satisfies this equation.

Checking the possible integer values of x , we find:

$$x = \pm 1 \quad \Rightarrow \quad x^2 = 1 \quad (\text{not } 3).$$

$$x = \pm 2 \quad \Rightarrow \quad x^2 = 4 \quad (\text{not } 3).$$

$$x = \pm 3 \quad \Rightarrow \quad x^2 = 9 \quad (\text{not } 3).$$

Clearly, no integer satisfies $x^2 = 3$.

Therefore, there is no integer x such that $x^2 = 3$.

Proof by Cases

Definition: Proof by cases is a technique where we divide a proof into different cases and prove the statement separately for each case. It is used when a statement can be true in several different ways, each of which needs to be verified.

Procedure:

- ▶ Identify all possible cases.
- ▶ Prove the statement for each case.
- ▶ Conclude the proof after all cases are covered.

When to Use Proof by Cases

Use proof by cases when:

- ▶ There are multiple possible scenarios for the statement.
- ▶ The conditions for the statement change depending on different situations.
- ▶ A direct proof is complicated or infeasible.

Cases: Example 1

Theorem: Any integer n is either even or odd.

Proof: We will prove this by considering two cases.

Case 1: n is even.

By definition, if n is even, then there exists an integer k such that:

$$n = 2k.$$

Since $n = 2k$, n is even by definition.

Case 2: n is odd.

If n is odd, then there exists an integer k such that:

$$n = 2k + 1.$$

Since $n = 2k + 1$, n is odd by definition.

Since every integer n is either even or odd, the statement is proved.

Cases: Example 2

Theorem: The sum of two integers is even if and only if both integers have the same parity (both even or both odd).

Proof: We will prove this by considering all possible cases for the parity of the integers.

Case 1: Both integers are even.

Let $a = 2m$ and $b = 2n$, where m and n are integers. The sum is:

$$a + b = 2m + 2n = 2(m + n),$$

which is even.

Case 2: Both integers are odd.

Let $a = 2m + 1$ and $b = 2n + 1$, where m and n are integers. The sum is:

$$a + b = (2m + 1) + (2n + 1) = 2(m + n + 1),$$

which is also even.

Cases: Example 3

Case 3: One integer is even and the other is odd.

Let $a = 2m$ and $b = 2n + 1$, where m and n are integers. The sum is:

$$a + b = 2m + (2n + 1) = 2(m + n) + 1,$$

which is odd.

Conclusion: The sum of two integers is even if and only if both integers have the same parity.

Proof by Cases: $|xy| = |x| \cdot |y|$

Theorem: Prove that for all real numbers x and y , we have the identity:

$$|xy| = |x| \cdot |y|.$$

Proof by Cases:

▶ **Case 1:** $x \geq 0$ and $y \geq 0$

▶ $|x| = x$ and $|y| = y$

▶ $|xy| = x \cdot y$

▶ $|x| \cdot |y| = x \cdot y$

▶ **Case 2:** $x \geq 0$ and $y < 0$

▶ $|x| = x$ and $|y| = -y$

▶ $|xy| = |x \cdot (-y)| = -x \cdot y$

▶ $|x| \cdot |y| = x \cdot (-y) = -x \cdot y$

Proof by Cases: $|xy| = |x| \cdot |y|$

► **Case 3:** $x < 0$ and $y \geq 0$

- $|x| = -x$ and $|y| = y$
- $|xy| = |-x \cdot y| = -x \cdot y$
- $|x| \cdot |y| = (-x) \cdot y = -x \cdot y$

► **Case 4:** $x < 0$ and $y < 0$

- $|x| = -x$ and $|y| = -y$
- $|xy| = |-x \cdot (-y)| = x \cdot y$
- $|x| \cdot |y| = (-x) \cdot (-y) = x \cdot y$

► **Case 5:** $x = 0$ or $y = 0$

- If $x = 0$, then $|xy| = 0$ and $|x| \cdot |y| = 0$.
- If $y = 0$, then $|xy| = 0$ and $|x| \cdot |y| = 0$.

Conclusion: In all cases, $|xy| = |x| \cdot |y|$. Therefore, the proof is complete.

Summary of Proof by Cases

- ▶ Proof by cases is used when there are multiple scenarios to consider.
- ▶ We divide the problem into separate cases and prove each one.
- ▶ Once all cases are proven, the overall statement is concluded.

Remember: Each case must cover all possible outcomes, and no case can be overlooked.